Student Management System

Department (Did (PK), Dname, Location)
Student (Sid(PK), Sname, Age, Thana, Distinct, Email, DID (FK))
Course (CCode (PK), Cname, Credit,DID (FK))
Result (Sid(FK),CCode (FK), Composite key (Sid,CCode), Mark)

-- Create the Department table

```
CREATE TABLE Department (
   Did INT PRIMARY KEY,
   Dname VARCHAR(100) NOT NULL,
   Location VARCHAR(100)
);
```

-- Create the Student table

```
CREATE TABLE Student (
   Sid INT PRIMARY KEY,
   Sname VARCHAR(100) NOT NULL,
   Age INT,
   Thana VARCHAR(100),
   District VARCHAR(100),
   Email VARCHAR(100) UNIQUE,
   DID INT,
   FOREIGN KEY (DID) REFERENCES Department(Did)
);
```

-- Create the Course table

```
CREATE TABLE Course (
   CCode varchar(16) PRIMARY KEY,
   Cname VARCHAR(100) NOT NULL,
   Credit INT,
   DID INT,
   FOREIGN KEY (DID) REFERENCES Department(Did)
);
```

-- Create the Result table

```
CREATE TABLE Result (
   Sid INT,
   CCode varchar(16),
   Mark INT,
   PRIMARY KEY (Sid, CCode),
   FOREIGN KEY (Sid) REFERENCES Student(Sid),
   FOREIGN KEY (CCode) REFERENCES Course(CCode)
);
```

-- Insert data into Department table

```
INSERT INTO Department (Did, Dname, Location) VALUES
(1, 'Computer Science', 'Building A'),
(2, 'Electrical Engineering', 'Building A'),
(3, 'Mechanical Engineering', 'Building B'),
(4, 'Civil Engineering', 'Building D'),
(5, 'Business Administration', 'Building D'),
(6, 'Mathematics', 'Building D'),
(7, 'Physics', 'Building B'),
(8, 'Chemistry', 'Building C'),
(9, 'Biology', 'Building C'),
(10, 'Architecture', 'Building A'),
(11, 'Philosophy', 'Building D'),
(12, 'Economics', 'Building D'),
(13, 'Law', 'Building A'),
(14, 'Psychology', 'Building C'),
(15, 'History', 'Building B');
```

-- Insert data into Student table

```
INSERT INTO Student (Sid, Sname, Age, Thana, District, Email, DID) VALUES
(1, 'Alice', 20, 'Thana1', 'District1', 'alice@example.com', 1),
(2, 'Bob', 21, 'Thana2', 'District2', 'bob@example.com', 2),
(3, 'Charlie', 22, 'Thana3', 'District3', 'charlie@example.com', 3),
(4, 'David', 23, 'Thana4', 'District4', 'david@example.com', 1),
(5, 'Eve', 24, 'Thana5', 'District5', 'eve@example.com', 2),
(6, 'Frank', 20, 'Thana6', 'District6', 'frank@example.com', 3),
(7, 'Grace', 21, 'Thana7', 'District7', 'grace@example.com', 7),
(8, 'Hannah', 22, 'Thana8', 'District8', 'hannah@example.com', 8),
(9, 'Ivy', 23, 'Thana9', 'District9', 'ivy@example.com', 8),
(10, 'Jack', 24, 'Thana10', 'District10', 'jack@example.com', 10),
(11, 'Kathy', 20, 'Thana11', 'District11', 'kathy@example.com', 10),
```

(12, 'Leo', 21, 'Thana12', 'District12', 'leo@example.com', 12),
(13, 'Mona', 22, 'Thana13', 'District13', 'mona@example.com', 12),
(14, 'Nathan', 23, 'Thana14', 'District14', 'nathan@example.com', 14),
(15, 'Olivia', 24, 'Thana15', 'District15', 'olivia@example.com', 15);

-- Insert data into Course table

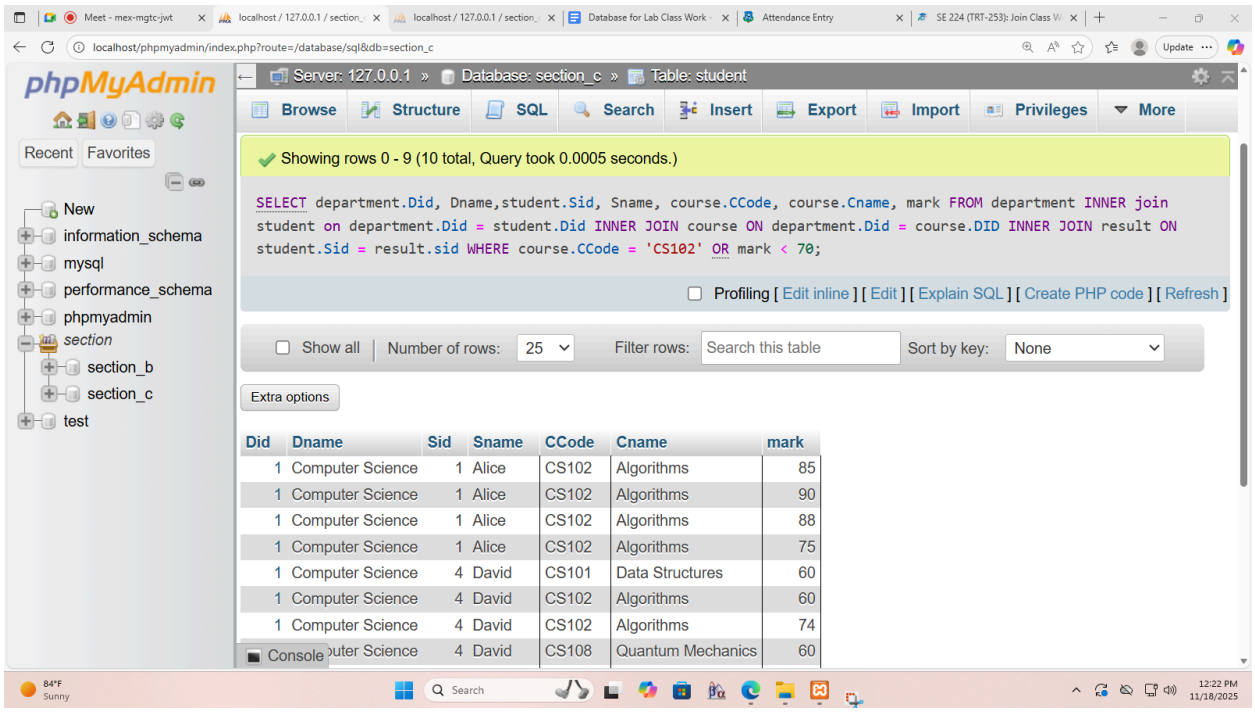
INSERT INTO Course (CCode, Cname, Credit, DID) VALUES
('CS101', 'Data Structures', 3, 1),
('CS102', 'Algorithms', 4, 1),
('EE103', 'Circuits', 4, 2),
('EE104', 'Thermodynamics', 3, 2),
('BA105', 'Statics', 3, 5),
('BA106', 'Accounting', 3, 5),
('BA107', 'Linear Algebra', 4, 5),
('CS108', 'Quantum Mechanics', 3, 1),
('CH109', 'Organic Chemistry', 2, 8),
('BIO110', 'Genetics', 2, 9),
('BIO111', 'Design Principles', 2, 9),
('CS112', 'Ethics in Law', 3, 1),
('EE113', 'Psychological Testing', 3, 2),
('PSY114', 'Macroeconomics', 3, 14),
('PSY115', 'Modern History', 2, 14);

-- Insert data into Result table

INSERT INTO Result (Sid, CCode, Mark) VALUES
(1, 'CS101', 85), (1, 'CS102', 90), (1, 'CS112', 75), (1, 'CS108', 88), (2, 'EE103', 80), (2, 'EE104', 82), (2, 'CS101', 87), (3, 'CS101', 70), (3, 'CS102', 78), (3, 'CH109', 65), (3, 'BIO110', 72), (4, 'CS108', 60), (4, 'EE104', 74), (5, 'CS112', 93), (5, 'PSY114', 88), (5, 'PSY115', 85), (5, 'BIO111', 89), (6, 'CS102', 81), (6, 'BIO110', 78), (6, 'CH109', 70), (7, 'CS101', 90), (7, 'CS112', 95), (7, 'EE103', 85), (7, 'BA105', 88), (8, 'CH109', 62),(9, 'BA106', 75), (9, 'BIO111', 82), (9, 'CS112', 90), (9, 'PSY115', 78), (10, 'CS101', 88), (10, 'CS102', 84), (10, 'EE113', 77), (10, 'BA105', 73);

Home Work

1. Retrieve a list of students and the names of the departments they belong to.
2. Retrieve all students and their department names, including students who do not belong to any department.
3. Retrieve all departments and the names of students in each department, including departments with no students.
4. Retrieve all students and departments, showing department names for students and students for departments, including rows without a match.
5. Join and output of all 4 tables
6. Retrieve a list of students, their course names, and department names for students who have taken courses.
7. Retrieve the names of students and the courses they have taken where the course credit is greater than 3.
8. Retrieve all students and their course names, including students who have not taken any courses, where the student's age is greater than 20.
9. Retrieve all departments and the students assigned to them, but only for departments located in "Building B".
10. Retrieve all students and departments, including unmatched rows, where the department name contains "Tech".
11. Retrieve a list of students and the names of the departments they belong to.



Did	Dname	Sid	Sname	CCode	Cname	mark
1	Computer Science	1	Alice	CS102	Algorithms	85
1	Computer Science	1	Alice	CS102	Algorithms	90
1	Computer Science	1	Alice	CS102	Algorithms	88
1	Computer Science	1	Alice	CS102	Algorithms	75
1	Computer Science	4	David	CS101	Data Structures	60
1	Computer Science	4	David	CS102	Algorithms	60
1	Computer Science	4	David	CS102	Algorithms	74
1	Computer Science	4	David	CS108	Quantum Mechanics	60